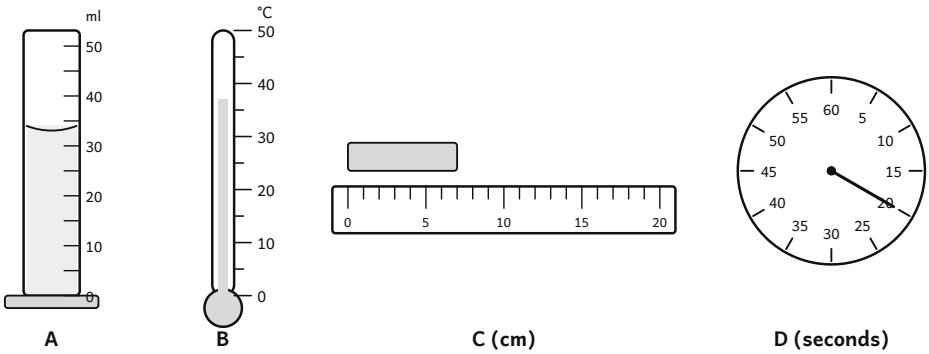
WORKING SCIENTIFICALLY

The Scientist's Toolkit

Questions, enquiry types, fair tests, measuring, tables and graphs, patterns, conclusions and models



A B C (cm) D (seconds)

In this pack

- **Learn it:** Thinking and working like a scientist ★☆☆
- **Fact check:** Tools of the scientist ★☆☆
- **Think it through:** Fair tests, graphs and results ★★☆
- **Investigate:** Which fabric soaks up the most water? ★★☆
- **Real-world case study:** Two Boses: scientists from Bengal ★★★
- **Review and challenge:** Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · W1 · WORKING SCIENTIFICALLY

Thinking and working like a scientist

Name: Class: Date:

Learning goals

- Ask questions that can be tested and choose a type of enquiry.
- Plan a fair test, measure accurately and record results in tables and graphs.
- Spot patterns and anomalies, draw conclusions and use models.

Scientists are curious people who ask questions about the world and then **collect evidence** to answer them. You already work like a scientist whenever you wonder "What will happen if...?" and then try it to find out. This toolkit is used in **every** science topic you study this year.

1. Ask a question

A good scientific question can be **answered by collecting evidence**. "Does warm water dissolve sugar faster than cold water?" can be tested. "Is sugar nicer than salt?" is an opinion, so it cannot be tested scientifically.

2. Choose a type of enquiry

Type of enquiry	Example from Grade 5
fair test	Does the length of a ruler change the pitch of its twang?
observing over time	How does a bean seed change as it germinates?
pattern seeking	Do people with longer legs jump further?
identifying and classifying	Which materials are transparent, translucent or opaque?
research (secondary sources)	How did Ole Rømer find out that light takes time to travel?

3. Plan a fair test

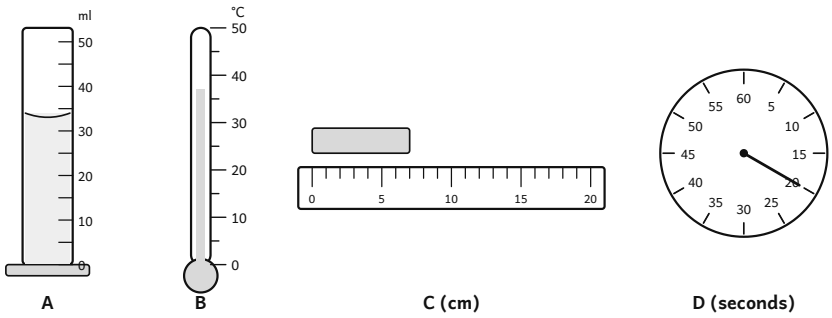
In a **fair test** you change **one** thing, measure **one** thing and keep **everything else the same**. These things are called **variables**. If you change two things at once, you cannot tell which one caused the result.

Change (independent variable)	the one thing you choose to change
Measure (dependent variable)	what you measure or observe to see the effect
Keep the same (control variables)	everything else that could affect the result

Write a **prediction** with a reason: "I think ... will happen **because** ...". Your reason should use science you know.

4. Measure accurately

Measuring...	Equipment	Unit
length	ruler, tape measure	mm, cm, m
mass	balance (scales)	g, kg
volume	measuring cylinder	ml, l
time	stopwatch	s, min
temperature	thermometer	°C
force	forcemeter	N (newtons)
light / sound	light meter, sound meter	lux, dB



Read each scale carefully, with your eye level with the mark.

Repeat measurements (often three times) and work out the **mean** (average) to make results more **reliable**. A result that does not fit the pattern is called an **anomaly** (an odd result). Check it and measure again.

> **Remember**
> To find the **mean**: add up the results, then divide by how many results there are. (20 + 22 + 24) ÷ 3 = 66 ÷ 3 = **22**.

5. Record and present

- A **table** has clear headings with **units**, e.g. Time (s).
- Use a **bar chart** when the thing you changed is in **groups or words** (types of soil, kinds of fabric).
- Use a **line graph** when the thing you changed is a **number that changes smoothly** (time, temperature, distance).

6. Find patterns and conclude

Look for a **pattern**: "The **longer** the ruler, the **lower** the pitch." Then write a **conclusion** that answers your question, using the data as evidence, and say whether it **supports your prediction**. Finally, **evaluate**: How could you improve the investigation?

7. Use models

Scientists use **models** to explain things that are too small, too big, too fast or too slow to see: the particle model of solids, liquids and gases; a globe and torch for day and night; a wave diagram for sound. Every model is useful, but every model has **limitations**: it is never exactly like the real thing.

> **Key words**

variable	something that can change in an investigation
fair test	only one variable is changed
prediction	what you think will happen, with a reason
reliable	results you can trust because they are repeated
mean	the average of a set of results
anomaly	a result that does not fit the pattern
conclusion	what the results show, answering the question
model	a way of representing an idea or object

FACT CHECK · W1 · WORKING SCIENTIFICALLY

Tools of the scientist

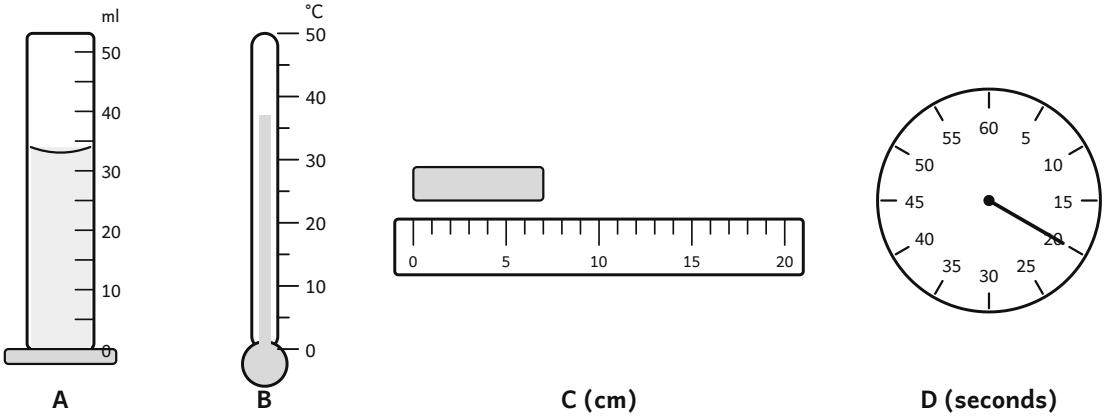
★☆☆
Grade 5

Name: Class: Date:

Learning goals

- Read scales accurately.
- Match units to measurements.
- Name types of enquiry.

1 Read each instrument. Include the unit. [4]



A B C (cm) D (seconds)

Instrument	A measuring cylinder	B thermometer	C ruler	D stopwatch
Reading				

2 Match each measurement to its unit. [6]

Draw a line to match each pair, or write the letter.

mass of an apple ●	● **A** decibels (dB)
force of a push ●	● **B** lux
loudness of a horn ●	● **C** seconds (s)
brightness of a room ●	● **D** millilitres (ml)
volume of juice ●	● **E** newtons (N)
time for a race ●	● **F** grams (g)

3 **Sort the questions.** [6]

SORT THESE

Which fabric dries fastest in the sun? **Does salt make ice melt faster?**
Is red the best colour? **Which song is the most beautiful?**
Do bigger seeds grow taller plants? **Are cats nicer than dogs?**

Can be tested scientifically	Cannot be tested (opinion)

4 **Write the type of enquiry: fair test, observing over time, pattern seeking, identifying and classifying or research.** [5]

Investigation	Type of enquiry
Recording the height of a sunflower every week	
Sorting leaves by their shape	
Testing which paper towel soaks up most water	
Reading about how vaccines were discovered	
Do children who sleep more have faster reaction times?	

Fair tests, graphs and results

★★☆ Grade 5

Name: Class: Date:

Learning goals

- Spot unfair tests.
- Choose the right graph.
- Calculate a mean and spot anomalies.

1 **Rima wants to find out whether ice melts faster in salt water. She puts one ice cube in a cup of salt water in the sun and a bigger ice cube in a cup of tap water in the shade.** [4]

a) Why is this not a fair test?

b) How should she change her plan?

2 **Bar chart or line graph? Tick the best choice.** [4]

Results	Bar chart	Line graph
Height of a plant measured every day for two weeks		
Number of learners who like each fruit		
Temperature of cooling tea every 5 minutes		
Water held by sand, soil and clay		

3 **Tanvir timed how long a paper helicopter took to fall three times.** [3]

Try	1	2	3	Mean
Time (s)	2.1	2.3	2.2	

a) Work out the mean. Show your working.

b) Why is it better to repeat the measurement?

4 **Write a prediction with a reason for this question: "Will a wet towel dry faster on a windy day or a still day?"** [2]

INVESTIGATE · W1 · WORKING SCIENTIFICALLY

Which fabric soaks up the most water?

★★☆ Grade 5

Name: Class: Date:

Learning goals

- Plan a fair test with repeats.
- Identify an anomaly and calculate means.
- Present and evaluate results.

Class 5D wanted to find the best fabric for a cleaning cloth. They dipped a 10 cm × 10 cm square of each fabric in water for 30 seconds, let it drip for 10 seconds, then squeezed it into a measuring cylinder. They repeated each test three times.

1 **Complete the plan.** [4]

Question	
Change	
Measure	
Keep the same	

2 **Here are the results.** [5]

Fabric	Test 1 (ml)	Test 2 (ml)	Test 3 (ml)	Mean (ml)
cotton gamchha	18	19	17	
jute (*pat*)	12	25	13	
polyester	6	5	7	
newspaper	9	8	10	

a) Circle the anomaly. Suggest a reason for it.

.....

.....

b) Work out the mean for each fabric and write it in the table. Leave out the anomaly.

3 **Draw a bar chart of the means.** [3]



4 **Write *a* conclusion *and* suggest one improvement.** [3]

.....

.....

Two Boses: scientists from Bengal

Name: Class: Date:

Learning goals

- Learn how real scientists work.
- Link evidence, creativity and new instruments.
- Reflect on sharing knowledge.

Science · History · Maths · Language · Ethics

Jagadish Chandra Bose was born in **1858** in Mymensingh, in what is now Bangladesh. He studied in Kolkata, Cambridge and London, then became a professor in Kolkata. In **1895** he showed that invisible radio waves could ring a bell and set off a small explosion in another room, one of the earliest demonstrations of **wireless** signals.

Later he asked a bold question: do **plants** respond to the world as animals do? Plant growth is far too slow to see, so he invented the **crescograph**, which magnified tiny movements about **10 000 times**. His careful measurements showed that plants react to light, heat, touch and chemicals. Bose believed knowledge should be **shared** with everyone and was reluctant to make money from patents. He also wrote one of the first science fiction stories in Bengali, about a cyclone stopped by a bottle of hair oil!

Satyendra Nath Bose taught physics at **Dhaka University** from 1921. In **1924**, in Dhaka, he wrote a new way of explaining light and sent it to **Albert Einstein**, who was so impressed that he translated it into German himself. A whole family of tiny particles, **bosons**, is named after him.

Based on biographies of J. C. Bose (Bose Institute) and S. N. Bose (University of Dhaka).

1 **Put these events in order.** [2]

- ☐ S. N. Bose sends his paper to Einstein (1924).
- ☐ J. C. Bose demonstrates wireless signals (1895).
- ☐ J. C. Bose is born in Mymensingh (1858).
- ☐ S. N. Bose starts teaching at Dhaka University (1921).

2 **Maths. A plant grows 0.002 mm in one minute. The crescograph magnifies 10 000 times. How far does the pointer move in one minute?** [2]

.....

3 **Why did J. C. Bose need to invent a new instrument? What does this tell you about how science moves forward?** [2]

.....

.....

.....

4 **J. C. Bose shared his discoveries freely instead of keeping them secret to make money. What are the benefits and problems of sharing?** [2]

Benefits of sharing	Problems of sharing

5 **Design. Invent an instrument to measure something too small, slow or fast to see. Draw it and explain what it measures.** [2]

My instrument

Show what you know

★★★ Grade 5

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

M	V	Y	L	P	R	E	D	I	C	T
T	U	A	Y	L	R	D	K	O	C	G
P	W	I	R	G	W	V	W	T	G	R
A	Y	U	P	I	L	C	A	C	M	A
T	I	S	B	F	A	W	S	F	E	P
T	V	Y	A	A	L	B	D	R	A	H
E	L	A	A	I	A	G	L	H	S	O
R	V	C	S	R	U	V	B	E	U	P
N	D	T	A	B	L	E	H	U	R	R
L	H	M	O	D	E	L	W	P	E	G
O	H	P	B	R	E	P	E	A	T	A

fair **graph** **measure** **model** **pattern** **predict** **repeat** **table** **variable**

2 **Which unit measures force?** [1]

- ◯ grams
- ◯ newtons
- ◯ litres
- ◯ degrees Celsius

3 **A result that does not fit the pattern is called...** [1]

- ◯ *a* conclusion
- ◯ *a* variable
- ◯ an anomaly
- ◯ *a* prediction

4 **Draw a results table for this investigation: "How does the temperature of water change the time sugar takes to dissolve?" Include headings with units and space for three repeats.** [3]

My table

5 **Challenge. We often draw particles in a solid as circles in neat rows. Give one way this model is useful and one limitation.** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
ask a question that can be tested	○	○	○
name the five types of enquiry	○	○	○
plan a fair test with variables	○	○	○
read scales and use the right units	○	○	○
calculate a mean and spot an anomaly	○	○	○
choose a bar chart or line graph	○	○	○
explain how J. C. Bose combined measurement and creativity	○	○	○